

$$f^{-1}(2)=a, f^{-1}(3)=b, f^{-1}(4)=c. \quad \square$$

Example 20: Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be such that $f(x) = x+1$. Is f invertible, and if it is, what is its inverse?

Solution: By the conclusions of Example 10 and Example 15 and from Definition 8, f is a bijective function. By Defn. 9, f is invertible.

Let, $f(x) = y$, where $y \in \mathbb{Z}$. $\therefore x+1 = y \Rightarrow \boxed{x = y-1}$.

$$f^{-1}(x+1) = x \Rightarrow f^{-1}(y-1+1) = y-1 \Rightarrow \boxed{f^{-1}(y) = y-1} \text{ for any arbitrary } y$$

This means that $y-1$ is the unique element of $\mathbb{Z}$ that is sent to y by f . $\square$

Example 21: Let f be the function from $\mathbb{R}$ to $\mathbb{R}$ with $f(x) = x^2$. Is f invertible?

Solution: We know that $\mathbb{Z} \subseteq \mathbb{R}$. ~~By Example 14, we know that~~

$\therefore 2, -2 \in \mathbb{R}$. $f(2) = 2^2 = 4 = (-2)^2 = f(-2)$. But $2 \neq -2$. $\therefore$ By Defn. 5, f is not one-to-one. By Defn. 8, f is not invertible. By Defn. 9, f has no inverse. $\square$

Example 22: Let f be the function from $\mathbb{R}^+$ to $\mathbb{R}^+$ with $f(x) = x^2$. Is f invertible?

Solution: Let, $x, y \in \mathbb{R}^+$ s.t. $f(x) = f(y)$. $\therefore x^2 = y^2 \Rightarrow x^2 - y^2 = 0$

$\Rightarrow (x+y)(x-y) = 0$. $\therefore x+y=0$ or $x-y=0 \Rightarrow x=-y$ or $x=y$. $\therefore x, y \in \mathbb{R}^+$, $x=-y$ is an invalid solution. $\therefore x=y$. By Defn. 5, f must be one-to-one.

Let, $f(x) = z$ where $z \in \mathbb{R}^+$. $\therefore x^2 = z \Rightarrow x = \pm\sqrt{z}$. $\therefore z \in \mathbb{R}^+$, $x = -\sqrt{z}$ is the only valid solution. We know that for any arbitrary $z \in \mathbb{R}^+$, $\sqrt{z} \in \mathbb{R}^+$. By Defn. 7, f must be onto.

By Defn. 8, f is bijective. From Defn. 9, f must have an inverse.

$$f^{-1}(x^2) = x \Rightarrow \boxed{f^{-1}(z) = \sqrt{z}} \quad \square$$

Definition 10: Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The **composition** of the functions f and g , denoted for all $a \in A$ by $f \circ g$, is the function from A to C defined by